

StocksDeepDive — Complete User Manual

stocksdeepdive.com · Version of 30 August 2026 · supersedes the 13 August 2026 edition

This manual describes every page, every table, every score and every colour on the public site, exactly as deployed. Nothing here is left to interpretation: each formula is written out, each threshold is stated, and each colour is defined. It replaces the 13 August 2026 edition, which had fallen behind the site in several places — most importantly, it described the entry-timing tools (Investment Signal, Trade Setup, buy/stop/target levels) as public features. They are not: on the live public site those are computed and used internally, but never shown. What a signed-out or signed-in (non-admin) visitor actually sees is a **Value Score** and a set of neutral, described calculations — no buy/sell verdicts. This edition documents the site exactly as a real visitor experiences it, with the owner's own admin-only tools broken out into a separate appendix (§12) rather than presented as public features.

1. Site structure and navigation

The site has ten Streamlit pages, plus a statically-served blog that sits outside the Streamlit app entirely:

URL	Page	Purpose
/	Home	Landing page: search, featured analysis, market strip, feature overview
/deep-dive	Stock Deep Dive	Complete analysis of one ticker
/comparison	Side-by-side Comparison	Two or more tickers on identical criteria
/scanner	Stock Scanner	Rank a whole index (ASX 200, S&P 500, etc.)
/research	Rational Compounder Analysis	The author's hand-built workbook research
/portfolio	My Portfolio	Your own tracked holdings (sign-in required)
/methodology	How the scores work	Plain-English methodology reference
/model-history	Model history	A changelog of Rational Compounder research rebuilds
/about	About	Who runs the site and why
/privacy	Privacy policy	What data the site handles
/blog, /blog/<slug>	Blog	Articles, served as static HTML — not a Streamlit page, and not part of the header nav (footer link only)

A tenth Streamlit page, `/blog-admin`, exists to write and manage blog posts — it is admin-only and excluded from search-engine indexing (`robots.txt`); it is not part of the public site and isn't covered further in this manual.

The search box is the primary navigation. It appears at the top of every page. The rule is simple and absolute: type one ticker and press Search (or Enter) for a Deep Dive; type two or more tickers — separated by spaces, commas or new lines — for a Comparison. ASX tickers use the `.AX` suffix (`CSL.AX`); US tickers use none (`AAPL`); the two can be mixed freely in one comparison. Duplicate tickers are ignored, case does not matter.

Below the search box sit four buttons, in this order: **Rational Compounder Analysis, Side-by-side Comparison, Stock Scanner, My Portfolio**. These open the pages directly — My Portfolio itself handles showing a sign-in prompt if you aren't signed in, rather than gating the click.

Example chips. On the home page and on Deep Dive/Comparison empty states, clickable chips run an example search with one click.

Shareable URLs. Results pages keep the address bar in sync with what's on screen — `stocksdeepdive.com/deep-dive?ticker=CSL.AX` opens directly into a live CSL analysis, and `stocksdeepdive.com/comparison?tickers=CSL.AX,BHP.AX` runs that comparison directly. Copying the address bar at any time captures the current view.

Clicking the StocksDeepDive logo returns to the home page from anywhere.

The "RC view" control, if you notice it in the top corner near Sign Out, is the site owner's own admin unlock (an access-key prompt) — it switches that one browser into the full admin view described in §12. It has no function for anyone without the key, and entering the wrong one simply shows "Incorrect key."

2. The home page

From top to bottom:

Market tape. A scrolling strip of live quotes (ASX 200, S&P 500, NASDAQ, AUD/USD, plus sample stocks) with today's percentage move, coloured green for a gain and red for a loss. If quotes can't be fetched, the tape simply doesn't render.

Account bar. Sign In when signed out; your name, a feedback button and Sign out when signed in. See §9 for what signing in adds and how sign-in actually works.

Hero and search. The headline, the search box with an Analyze button, the example chips, and the usage rule ("One ticker = full Deep Dive · Two or more = side-by-side Comparison").

Featured analysis card. A live, attention-lite Deep Dive of one stock, rotating daily through a fixed list and cached for several hours. It shows the Value Score on a colour-coded gauge, intrinsic value, margin of safety, Quality Score, and a price sparkline against the 50-day average. The button beneath opens the full Deep Dive.

Your watchlist (signed-in users with saved stocks only): a chip row of your saved tickers — one click re-runs any of them.

Market mood strip. Up to four tiles: AU and US market mood (Hopeful / Neutral / Anxious — a rolling reading of news tone for that country; informational only, it feeds no score) and index levels with today's move. A tile that can't be fetched is silently omitted and returns when the source recovers.

Feature cards, how-it-works, coverage cards and the launch banner complete the page.

3. The scoring engines (used identically everywhere)

Every tool on the site runs the same engine. Live data comes from Yahoo Finance (prices, volumes, fundamentals, analyst estimates), Google Trends (search interest), NewsAPI + Yahoo news (headlines) and StockTwits (social chatter); statement history depth depends on the data source in use — a handful of years (typically 4–5) from Yahoo by default, or up to about ten years when the site is additionally configured with an EODHD data key (not currently set on the production deployment, so live figures today run on the shorter Yahoo-only history).

3.1 Quality Score (0–100)

"Is this a good business?" — computed from the company's own fundamentals:

```
Quality = 50 + ROIC × 100 × 0.25 + ROE × 100 × 0.20 + Profit Margin × 100 × 0.15  
+ Revenue Growth × 100 × 0.15 + Earnings Growth × 100 × 0.15  
+ 10 if free cash flow positive (-5 if negative)  
- min(Debt/Equity × 0.05, 15) [debt penalty, capped]
```

- **ROIC** (return on invested capital) is the lead metric — profit per dollar of *total* capital (equity and debt), so a business can't look high-quality on ROE alone by borrowing heavily. Revenue and earnings growth are clamped to ±50% before weighting, so a +300% figure off a tiny base can't max the score.
- **Profitability gate**: a loss-making or cash-burning business (negative margin, net income or free cash flow) is capped at **55** regardless of growth.
- The final score is clamped to 0–100. If no fundamental data was available at all, the score defaults to 50 and is rendered in red (see the red-flag rule, §3.7).

3.2 Intrinsic value and Margin of Safety

"What is it worth?" The valuation ladder, in order:

1. **DCF** — used whenever free cash flow is positive. The base cash flow is the latest reported year, adjusted for average capital expenditure so one unusually heavy investment year doesn't collapse the estimate; if that figure still deviates by more than 40% from the median of the last few years, the median is used instead as a smoothing step.
2. **P/E blend** fallback when a DCF isn't possible — labelled as such in the "Val Method" fields.
3. **N/A** when neither works (typically financials, or no positive EPS/FCF).

The **growth rate** feeding the DCF isn't a simple priority list — where both an analyst 5-year consensus estimate and a "clean" multi-year historical FCF growth rate are available, the *lower* of the two is used (a conservative blend, not a straight pick of one over the other); otherwise whichever of the two is available is used alone; failing that, the company's own reported growth figures; failing that, a flagged default. Whatever the source, growth is capped by the company's own market-cap tier — roughly 8% for mega caps, rising to 20% for small caps — so a small, fast grower isn't artificially capped at a large company's pace, and a mega-cap outlier estimate can't run away with the model.

The **discount rate** is CAPM-based — the stock's own beta against its market's risk-free rate and equity risk premium — floored at 7.5% (and the beta feeding it floored at 0.6) and capped at 15%, since a measured cost of capital or beta below that usually reflects a data artefact rather than a genuinely low-risk business. Terminal growth is set by the stock's currency.

```
Margin of Safety (MOS) = (Intrinsic Value - Price) / Intrinsic Value × 100
```

Valuation label: MOS >= 25 -> **UNDERVALUED** · MOS < 0 -> **EXPENSIVE** · between -> **FAIR** · no intrinsic value -> **N/A**.

Per-stock DCF overrides (growth / discount / terminal) can be typed into the DCF Parameters table on Comparison and Scanner (§5, §6) by any visitor. Overrides live only in your own browser session, reset when you leave, and never affect any other visitor.

3.3 Psychology Score

"Which way is the crowd leaning?" — computed from 3 months of price behaviour:

```
Fear = (3-month High - Price) / High × 100 [distance below the recent high]
Greed = max((Price - MA50) / MA50 × 100, 0) [stretch above the 50-day average]
FOMO = Greed + max(weekly change, 0)
Psychology = Fear - Greed - FOMO
```

Higher = fearful (contrarian-friendly); lower = greedy/overheated. The Sentiment label reads directly off it: > 20 **FEARFUL** · > 5 **CALM** · < -20 **OVERHEATED** · < -5 **GREEDY** · else **NEUTRAL**.

3.4 Discovery Score

"Is the market noticing this stock?" — pure attention, deliberately containing no sentiment:

```
Discovery = Activity + Volume Ratio × 10 + Trend Score + News Score + Social Score
```

where Activity = |weekly change|, Volume Ratio = latest volume ÷ 3-month average volume, Trend = Google search interest, News = NewsAPI + Yahoo headline scoring, Social = StockTwits message volume and bull/bear balance. In **attention-lite** contexts (overnight scans, live scans over 100 tickers, the featured card and the weekly digest) the Trend/News/Social terms are skipped and Discovery reflects price and volume only — always labelled where it applies.

3.5 Moat Score (0–100) — new

Quality (above) measures how good the business is **right now**. Moat is a different question: how likely is it to **stay** that good, over time? It's shown on the Deep Dive page and, where a value has already been computed and stored, on the Side-by-side comparison and Stock Scanner tables — and, with `MOAT_IN_VALUE_SCORE` set (see §13), it is also folded directly into the Value Score below at a 15% weight. Four pillars, each computed from the company's own multi-year statement history:

Pillar	Weight	What it measures
Excess-return spread	30 pts	TTM return on capital (ROIC, or ROE for banks/insurers) minus the cost of that capital. <=0% spread scores 0, 0–5% scores 10, 5–15% scores 20, above 15% scores the full 30.
Persistence	25 pts	The share of available fiscal years in which return on capital cleared 12%. Capped at 20/25 while fewer than 8 years of statement history are on file.

Pillar	Weight	What it measures
Pricing power	25 pts	The gross-margin trend (falls back to operating margin, flagged, if no Gross Profit line is reported): held/expanded margin (10 pts), stability (10 pts), and margin holding up while revenue actually grew (5 pts).
Reinvestment	20 pts	Incremental return on newly-deployed capital, oldest available year to newest. A capital-light business that shrinks invested capital while holding or growing profit scores 16 here directly.

Above 70 = **Strong moat**, 40–70 = **Moderate moat**, 40 and below = **Weak/no moat**.

An **erosion overlay** runs independently of the score: if return on capital and operating margin have both fallen meaningfully below their own recent multi-year average, the read becomes a "moat watch" caption (no score penalty); if that holds across the two most recent multi-year windows in a row, the read becomes "eroding" and the score itself is capped at 50.

If a pillar can't be computed from the data on file, it's dropped and the remaining pillars reweighted to 100 — never defaulted to a neutral score. Funds/ETFs don't get a Moat Score at all ("N/A (fund)"); a ticker with fewer than two usable statement years also shows "N/A."

3.6 The Value Score (0–100)

The headline number on the public site. Every component is clamped to a fixed band first so no single factor can dominate. Which formula runs depends on the `MOAT_IN_VALUE_SCORE` environment variable (§13) — currently **set (on)**, so Moat is folded in:

```
With MOAT_IN_VALUE_SCORE set (current production formula):
Value Score = Quality × 0.25 + Moat × 0.15 + MOS clamped to ±50 × 0.25
+ Psychology clamped to ±50 × 0.20 + Discovery clamped to 0–100 × 0.15

Without it (the formula used before Moat was folded in, and the one that
still runs for any ticker with no Moat Score – funds/ETFs, or too little
statement history):
Value Score = Quality × 0.35 + MOS clamped to ±50 × 0.25
+ Psychology clamped to ±50 × 0.20 + Discovery clamped to 0–100 × 0.20
```

When a ticker has no Moat Score, its 15% weight isn't dropped to zero — it's redistributed proportionally across the other four factors, so the second formula above is exactly what runs for that ticker even while the switch is on. The clamps affect only the score's arithmetic; tables always display the true, unclamped MOS/Psychology/Discovery values. This is the same underlying calculation the site's own code calls the "Long Score" internally (computed there via a `mode="moat_blend"` argument when the switch is on), and the same number the site's owner sees paired with an Investment Signal label in the admin view — see §12. On the public site it is shown purely as a descriptive number ("Value Score: 48.9 — FAIR" and similar), with no buy/hold/avoid label attached.

3.7 The red-flag rule

Wherever a number rests on a default or average because real data was unavailable, it is rendered in **red** — in every table, on every page. A red Quality of 50 means "no fundamentals were available; this is the base assumption," not a computed result. An estimate is never displayed as a fact.

4. Stock Deep Dive (/deep-dive)

Search one ticker to reach it. From top to bottom:

1. **Headline metrics:** Price, Intrinsic Value, MOS, Value Score. 2. **Follow / Watchlist:** a follow toggle lets any reader — signed in or not — leave just an email address to be notified when this stock's Rational Compounder research changes (a lighter commitment than the full watchlist below); signed-in users additionally get an Add/Remove watchlist button for this stock. 3. **Price chart:** the last 6 months of closes, the 50-day moving average, and the 3-month high/low context that Psychology is computed from. 4. **The plain-English case:** Why Buy / Why Wait / Risks in three columns, plus a suggested holding period. 5. **Price vs Intrinsic Value** bar chart and the **Value Score gauge** with its threshold caption. 6. **"What's driving the Value Score"** — each factor's exact point contribution. 7. **Quality** (gauge + driving-terms chart, including the ROIC term), **Psychology** (gauge + Fear/Greed/FOMO chart), **Discovery** (gauge + attention-terms chart), and **Moat** (gauge + driving-pillars chart, with an erosion caption when it applies — see §3.5). 8. **Compounder View (auto)** — a collapsed research section mirroring the Rational Compounder Analysis page's own layout (Fundamentals, Value vs Book, Retained Earnings, Earnings Trends, Cost of Capital, Fair Value), but computed live from this ticker's own reported statements and price history instead of drawn from the author's hand-built workbook. Metrics are coloured against the same threshold bands the workbook research uses; anything resting on an estimate or fallback is flagged red, same convention as everywhere else on the site. "Fair Value" here uses the same preview gate as the workbook version (see §7).

A position-disclosure line (whether the site's author personally holds, has never held, or has closed a position in this stock) appears on Research-style pages — see §9.

5. Side-by-side Comparison (/comparison)

Search two or more tickers. Every stock is fetched and scored live, with a progress bar (a warning shows a realistic time estimate whenever more than 25 tickers were requested). All tables share one visual language:

- **Infill bars** — a value above a coloured bar whose fill width is the value and whose colour is its band (see the full colour reference, §10).
- **Verdict pills** — coloured labels: green = UNDERVALUED / FEARFUL / Uptrend; amber = FAIR / GREEDY / Ranging; red = EXPENSIVE / OVERHEATED / Downtrend; grey = CALM / NEUTRAL / N/A.
- **Price-relative colouring** — Intrinsic Value is green when above the current price, red when below.
- **Red values** — the red-flag rule (§3.7) overrides all other colouring.

The tables, in order:

5.1 Comparison preview — Ticker, Price, Value Score for every stock scanned.

5.2 Side-by-side comparison — the main table, one row per ticker in the order you entered them. Columns: Ticker · Type (pill) · Price · Intrinsic Value (green/red vs price) · MOS (bar) · Value Score (bar) · Quality (bar) · Psychology (bar) · Discovery (bar) · Moat (bar) · Valuation (pill) · Sentiment (pill) · Trend (pill). The Moat column

shows only a value already computed and stored by an earlier nightly scan or Deep Dive view of that ticker — it is never computed live in this table, since a fresh multi-year fundamentals fetch per ticker would be too slow for a page with ten or more names — and reads "N/A" until one exists, coloured red whenever the erosion overlay (§3.5) has fired for that stock.

5.3 Average score by stock type — shown only when 5+ stocks were scanned (a group average of 2–3 names says nothing).

5.4 Highest Value Score in this scan — a one-line caption naming the top-scoring stock in the batch, explicitly framed as a sort result, not a recommendation.

5.5 DCF Parameters (collapsed expander) — the valuation mechanics per stock: Ticker · Price · Intrinsic Value · IV/Price (green above 1.00×, red below) · Upside % · DCF Growth (red when defaulted) · Growth Governor (which rule set the growth figure) · Discount · Perpetual · Value Score (bar). Click **Enable manual override** to type your own growth/discount/terminal per stock — blank = automatic; overrides are session-only.

A closing line reports how many stocks were analysed and how long the scan took.

6. Stock Scanner (/scanner)

Answers the bigger question: "across a whole index, what should I even look at?"

Selection. Tick Australia and/or USA, then pick exactly one universe — each maps to one real index, never a blend:

Universe	Constituents sourced from
ASX 200	Wikipedia S&P/ASX 200 (fallback: a curated local list)
ASX 300	asx300list.com (fallback: ASX 200 live, then the local list)
S&P 500	Wikipedia (fallback: a static 10-ticker list)
Nasdaq 100	Wikipedia
Russell 2000	iShares IWM ETF holdings (fallback: the last successfully fetched list, saved to disk)
Small Caps (S&P 600)	Wikipedia

The source actually used is always shown under the pickers, so a scan on fallback data is never silently presented as live.

Sector filter with heat dots. The Sector dropdown decorates each sector with a coloured dot — hot / medium / cold, ranked by dollar-volume-weighted 12-month return relative to the other sectors in this universe. Computing it for a universe the first time takes a few minutes; after that it's cached for the day.

Overnight scans — the instant path. Every night the server scans each configured universe by itself and stores the ranked result, including a stored Moat value and erosion read for each ticker (see §3.5) computed as part of that same run. When one exists for the selected universe, an expander appears with the full colour-coded table: Ticker · Type · Price · Intrinsic Value · MOS · Value Score · Quality · Psychology · Discovery · Moat · Valuation · Trend. Overnight scans are attention-lite (§3.4) and marked with their computation time;

results older than 3 days are treated as expired.

Live scan. The Run Scan button scores the selected universe/sector live, with a progress bar and an honest up-front time estimate. Scans over 100 tickers automatically run attention-lite. Results render in exactly the layout of §5 — including the Moat column, which (as on Comparison) shows a stored value only, never computed live in this loop.

7. Rational Compounder Analysis (/research)

The site's hand-built research — not computed, written. Every chart, threshold and colour band comes from the author's own research workbook.

Pick a Stock — the list is not a fixed set of names; it is read from whatever the current research data file actually covers, and grows automatically as the author adds and rebuilds coverage for new companies (a caption states how many companies are covered today). A **Model history** page (§1) lets you browse every past rebuild of this data alongside the current live version, as a changelog — not a claim about the accuracy of any past or future output.

Pick a Section:

- **Fundamentals** — years of monthly share price against the author's own multi-year average line, colour-banded metric gauges (each with a collapsed "what this measures" explanation in the author's own words), other headline metrics as cards, and a share-price-growth-by- year chart.
- **Value vs Book** — Intrinsic Value ÷ Book Value by year modelled, colour-banded, with the average line.
- **Retained Earnings** — the "value created per \$ retained" test at several horizons: for every dollar of earnings kept, how much market value was created — plus dividend metrics.
- **Earnings Trends** — EPS by year, EPS growth by year, P/E by year with average reference lines, plus the sheet's other figures.
- **Cost of Capital** — WACC vs ROIC by period: the fundamental test of whether the business earns more on its capital than that capital costs.
- **Fair Value** (*preview gated — see below*) — four independent valuation methods side by side against the current price: trailing P/E, forward P/E, DCF, and a 10-year equity method, with the exact inputs behind each.
- **Company Potential** (*preview gated — see below*) — the author's own Buffett/Munger-style judgment: Low/Medium/High ratings on management, moat, risk, pricing power and more, quick yes/no checks, and the full written analysis grouped by theme.

The Deep Dive page's own "Compounder View (auto)" section (§4) mirrors the first six of these — Fundamentals through Fair Value — computed live for that one ticker instead of drawn from the hand-built workbook; it has no live equivalent of Company Potential, since that section is the author's own judgment rather than a calculation.

A **position-disclosure line** appears here for the selected stock: whether the site's author personally holds it, has never held it, or has previously held and closed the position. This is the author's own portfolio disclosure, shown identically to every visitor — it is not a data signal and doesn't change with any score on the page.

On the "gated" sections: the site is currently free (see §9) — Fair Value and Company Potential are marked as preview content that will move behind a subscription if and when one opens, but nothing is actually restricted

today.

8. My Portfolio (/portfolio) — new since the last edition

Sign in to use this page — it shows a sign-in prompt otherwise. Four tabs track stocks you've told the site you hold:

- **Holdings** — your tracked positions.
- **Overview & P/L** — profit/loss summary across your holdings.
- **Health & News** — a health read and recent news for what you hold.
- **Progress** — how your tracked portfolio has moved over time.

This is entirely your own data, separate from the site author's own position-disclosure line described in §7/§9 — the two are unrelated features that happen to share the word "position."

9. Accounts, watchlist, follow and the weekly digest

Signing in takes one of two forms, offered side by side in the same Sign In popover:

- **Google sign-in** — one click, no password to create; the site receives only your name and email.
- **Email sign-in code** — type your email, the site emails a 6-digit code (via Mailgun) that expires after 15 minutes; five wrong attempts burns the code and a fresh one is needed. A signed-in session from either method lasts 90 days in your browser. Code emails are rate-limited (a handful per address per day) to stop the sign-in flow being used to spam an inbox.

Once signed in, both methods work identically everywhere on the site — there's no feature difference between a Google session and an email-code session.

Signing in unlocks:

- **The watchlist** — a personal saved-ticker list (an Add button on any Deep Dive); your saved stocks appear as one-click chips on the home page, and can be bulk-imported by pasting a list of tickers (up to 50 at a time).
- **Feedback** attributed to your email.
- **The weekly digest**, which arrives Monday mornings if your watchlist has stocks in it: each saved stock re-scored (attention-lite) with price, intrinsic value, MOS, Value Score, each ticker linking to its live Deep Dive. Remove all stocks from your watchlist to stop receiving it.

Follow is a separate, lighter-weight feature from the watchlist, and doesn't require a full sign-in: on the Research and Deep Dive pages, a Follow toggle lets you leave just an email address to be notified when that company's Rational Compounder research is updated (or, via a follow-all option, when any company's research changes) — it's a notification subscription for one thing changing, not a saved list you manage.

Feedback — the "Tell us what you think" button on any results page goes straight to the author's inbox, or email rationalcompounder@stocksdeepdive.com.

Subscriptions are not active — the site is currently free. There is no paywall or pricing page anywhere on the site today; everything described in this manual, including the sections marked "preview gated" in §7, is

available to every visitor right now.

10. Colour reference (public site)

Element	Green	Amber	Red	Grey
Value Score bar	> 50	30–50	<= 30	—
Quality bar	> 80	40–80	<= 40	—
MOS bar	> 25%	0–25%	<= 0%	N/A
Psychology bar	> 20	-5 to 20	<= -5	—
Discovery bar	> 50	25–50	<= 25	—
Moat bar	> 70	40–70	<= 40	N/A (fund/insufficient data)
Valuation pill	UNDERVALUED	FAIR	EXPENSIVE	N/A
Sentiment pill	FEARFUL	GREEDY	OVERHEATED	CALM / NEUTRAL
Trend pill	Uptrend	Ranging	Downtrend	—
Intrinsic value	above current price	—	below current price	N/A
IV/Price	> 1.00×	—	< 1.00×	N/A
Upside %	positive	—	negative	N/A
Any value in red bold	—	—	estimated/default input (red-flag rule)	—

A Moat value in red bold specifically means the erosion overlay (\$3.5) has fired for that stock ("watch" or "eroding"), on top of whatever colour band its score falls in — the same red-flag convention used for every other estimated/flagged value on the site, not a separate signal.

11. Data sources and honest limitations

Prices, fundamentals, and analyst estimates come from Yahoo Finance; search interest from Google Trends; headlines from NewsAPI and Yahoo; social chatter from StockTwits; country mood from GDELT. These are free public feeds: data can be delayed, revised, or occasionally missing — which is exactly what the red-flag rule exists to expose rather than hide. Statement history depth is typically 4–5 years (Yahoo Finance); the site can be configured with a deeper data source for up to about ten years, but that is not enabled on the current production deployment. Intrinsic values are estimates resting on stated assumptions. Nothing on the site considers any individual's circumstances, and nothing in it is financial advice — see the disclaimer in the site footer on every page.

12. Owner's view (appendix — not part of the public site)

Everything in this appendix is visible only to the site's owner, after unlocking "RC view" (§1) with the admin access key, or via a matching `?admin=` link. A public or signed-in visitor never sees any of this. It's included here for completeness, and so this manual doesn't silently omit what the "Value Score" number actually drives internally.

Investment Signal (the admin name for the Value Score band): above 70 -> **STRONG LONG** · above 50 -> **LONG** · above 30 -> **WATCHLIST** · otherwise **AVOID**. If no intrinsic value could be computed, the signal is capped at WATCHLIST.

Trade Setup — the timing axis. *"Is right now a sane entry?"* — deliberately separate from the Investment Signal:

```
Entry Zone = max(20-day support, MA50) — buy zone reached when Price <= zone × 1.05
Stop Loss = support-based
Targets: T1 = 20-day resistance, T2 = 60-day resistance, T3 = T2 × 1.10
Risk = Entry - Stop RRn = (Target n - Entry) / Risk
```

BUY requires all of: not in a confirmed downtrend, Psychology > 0, Discovery > 0, Price <= MA50 × 1.05, price near the Entry Zone, and RR1 >= 1.5. Otherwise WATCHLIST or AVOID. The **Trade Setup Score** (0–100) is a weighted display of the same gates (Trend Safety 20, Near Entry Zone 20, Risk/Reward 20, Price vs MA50 15, Psychology Momentum 12.5, Discovery Momentum 12.5), coloured green above 65, amber 45–65, red at or below 45.

Deep Dive (admin view) additionally shows the full Trade Setup section — gauge, gate-check chart, entry/stop/targets bar chart, and a caption on whether price is currently inside the entry zone.

Comparison and Scanner (admin view) additionally show: the Investment Signal and Trade Setup columns/pills throughout; a "Top Investment Candidate" block with the full written thesis and an "Opportunity Details" expander for the top five results; a collapsed **Trade Setup** table (Entry Zone, Stop Loss, Targets 1–3, Risk, RR1–3 coloured by rank across the rows on screen, an Early Exit watch flag) with a "Position management & early-exit rules" note; and, on Comparison only, an **Import screen CSV (admin)** panel that accepts a TradingView screener CSV export, scans it in batches of 25 tickers on demand, and shows each imported batch's results in its own isolated table — deliberately never merged into the main Scanner results.

13. Operations reference (site owner)

Configured with Railway environment variables; changing one restarts the service.

Variable	Set on production today?	Purpose
ADMIN_REFRESH_KEY	yes	The "RC view" / <code>?admin=</code> access key (§1, §12), and enables the Research page's workbook-rebuild admin panel
PAYWALL_ENABLED	set (see §9 — the live site currently shows no paywall/subscribe UI)	Master switch for the Stripe subscription flow; needs the Stripe and Google OAuth variables below

Variable	Set on production today?	Purpose
GOOGLE_OAUTH_CLIENT_ID / GOOGLE_OAUTH_CLIENT_SECRET	yes	Google sign-in (§9)
AUTH_COOKIE_SECRET	yes	Signs the sign-in session cookie (both Google and email-code sessions)
STRIPE_SECRET_KEY / STRIPE_PRICE_ID	yes	Subscription billing, inert while <code>PAYWALL_ENABLED</code> is off
MAILGUN_API_KEY / MAILGUN_DOMAIN	yes	Sends the email sign-in code and the weekly digest
FEEDBACK_SMTP_EMAIL / FEEDBACK_SMTP_APP_PASSWORD / FEEDBACK_TO_EMAIL	yes	Feedback-button email delivery
ANTHROPIC_API_KEY	yes	Optional live grammar/wording check on Company Potential free text
INDEXABLE_PAGES	yes	Controls which pages are allowed in <code>robots.txt</code> /sitemap
NIGHTLY_UNIVERSES	yes (ASX 200)	Comma-separated universes to pre-scan nightly (names exactly as in §6)
NIGHTLY_SCAN_UTC_HOUR	yes	Scan hour, UTC. Scans retry up to 3×/day if a run was interrupted
RAILWAY_VOLUME_MOUNT_PATH	yes (set by Railway)	Where persistent data — watchlists, overnight scans, compounder data, the Moat cache, scheduler state — actually lives, and survives redeploys
EODHD_API_KEY	not set	Optional deeper statement history (§3, §11); without it the site runs on Yahoo Finance's shorter history only
MOAT_IN_VALUE_SCORE	set to 1 (on)	Folds the Moat Score into the Value Score's own weighting — Quality 25% / Moat 15% / MOS 25% / Psychology 20% / Discovery 15%, proportionally reweighted to the pre-Moat formula for any ticker with no Moat Score (funds/ETFs, thin statement history). Read once at process start (<code>moat_engine.py</code>), so changing it restarts the service, same as any other variable here. See §3.5–§3.6 for the full formula.
SCHEDULER_ENABLED	code default <code>true</code>	Set to <code>false</code> to disable both the nightly scan and the weekly digest jobs
DIGEST_UTC_WEEKDAY / DIGEST_UTC_HOUR	code default (Sunday 21:00 UTC)	Weekly digest schedule

Variable	Set on production today?	Purpose
NEWS_API_KEY	optional	Enriches the News Score; without it, Yahoo headlines only

Manual commands from the service shell: `python nightly_scan.py "S&P 500"` (one universe now) and `python digest_engine.py` (send the digest now).